

CHILDREN & YOUTHS



THIS CHAPTER DESCRIBES exercises for children and youths. These exercises are designed to keep a large group active, while the coach can also follow-up individually with each climber. The chapter starts with a few thoughts on warming up before presenting examples of play-based climbing exercises you can use as a coach. Finally, more specific technical and physical exercises are presented. These are more appropriate for slightly older and more experienced children and youths.

The exercises in this chapter are a selection of the best exercises used by clubs all over Norway. We want to especially thank Oslo Climbing Club, Kolsås Climbing Club and Trondheim Climbing Club for their valuable contributions.

WARMING UP FOR CHILDREN

PHYSIOLOGICALLY, ADULT BODIES will start to prepare for an activity before getting started, through an increase in heart rate and body temperature, among other things. These changes in the transition from rest to activity are not seen in prepubescent children, and the warm-up must therefore be set up differently and last longer for children than for adults. Ideally, we want to gradually increase the level of activity, divided into short intervals of activity and rest, for a total period of 30 to 35 minutes. Since the warm-up will be a major part of the training session itself, it's important that it contains both fun and educational elements.



AREA OF FOCUS

As a coach, it's important that you have an area of focus for the session, and that all the parts of the session reflect this. If you're focusing on footwork, the general part of the warm-up should, more so than normally, activate the larger muscles in the legs. In the main part, you should choose different games and technique exercises that focus on different elements of footwork. To finish, you can gather the children and talk about what they have learned.

As with adults we start with a [general warm-up](#), before moving on to a more climbing-specific part. The general part should be play based and consist of exercises that challenge both physical and mental skills, as well as coordination. Games like 'Simon says', 'red light, green light' or 'crocodile challenge' are always popular, and you can be creative and adapt the exercises to your group. In the climbing-specific part, we preserve the playfulness while introducing exercises that primarily encourage activity on the wall but also focus on technical elements like footwork, weight transfer and balance.

CLIMBING GAMES

HERE ARE SOME examples of climbing games. Mix them up to stimulate variation in the training, but return to some of them on a regular basis. Your group is guaranteed to have a favourite, and kids like repetition and predictability.



SHARK ATTACK

All the children start sitting on the mats at an equal distance from the wall. When you call out 'shark attack', everybody should climb on to the wall as quickly as possible. The last one to leave the floor gets eaten by the shark.



Make up variations around the same theme. For example, a tsunami could mean they have to climb higher up the wall, or a storm might mean everybody has to get back down on to the mats again.

THE KING COMMANDS

All the children should start on the wall, preferably on a slab or a vertical section. They now have to do exactly what you tell them to do. If your command is to move their right hand, they have to do so, and if your command is to touch their right foot with their left hand, they have to do so. Your only limitation is your own imagination. You can adjust the level of difficulty by placing the children in different locations.



Be inventive and add technique training to the exercises. For example, have the children look at the footholds and hover their foot above the foothold for 2 seconds to train footwork precision. You can also command them to only use sidepulls in order to train balance.



THE FLOOR IS LAVA

The floor (or matting) is lava, and the children have to traverse from a defined starting point to a defined target, without falling into the lava. You can add a few safe 'islands' where they can rest before continuing climbing so that everybody can experience a feeling of mastery on their own terms.



Prepare the session in advance by taping a start and a finish. For the islands, you can use hula hoops or something similar on the ground.



PASSING

Divide the group into two and have one half start at the right-hand end of the wall and climb to the left. The other group starts at the opposite end and climbs to the right. They now have to pass each other on the wall. It's important that they don't climb above or below one other, as this might cause dangerous fall situations.



To make it extra challenging, tape an upper and a lower limit they have to stay within. You can also tape holds or zones where they are allowed to pass each other. This forces the children to plan and work together to solve the task.



TARGET PRACTICE

Each child gets a ball or a beanbag and has to climb on to the wall and aim for a target on the ground. A rope bag can be a suitable target. When they have thrown the ball they have to climb back down, retrieve the ball, climb back up and throw again. Give the group 1 minute per round and let them count how many times they can hit the target. This is a good warm-up exercise that forces the children to seek out good positions on the wall so they are able to throw accurately.



The children can start in corners where it is easier to find positions that allow them to let go with both of their hands. As they progress, they can have a go at more advanced positions.



PASS IT ON

Have the children spread out on the wall, about one metre apart. Whoever's furthest to the left is given an object that is to be passed on to the next child to the right. The object is then passed on from child to child all the way to the right, and then back again.



To make it more challenging, the children can spread further apart so that the object has to be thrown.



LIMBO

Have the children traverse a wall with a lot of holds while you're holding a broom handle against the wall for the children to climb under without touching the broom or the ground. The shorter the climber, the easier it is, so this is an exercise which can be extra fun if you participate yourself, because nothing is more fun for kids than being better than their coach! To make it extra challenging you can assign multiple coaches to hold brooms at different heights to make an obstacle course on the wall.



Try using a hula hoop for the children to climb through.



OCTOPUS

Have the children get established on the wall with both hands on a good hold and only one foot on a foothold. They should now try to touch as many holds as they can with their other foot for 15 to 30 seconds, before switching feet. Have them work in pairs, so that they can help each other count.



Positive handholds will allow them to move their hips away from the wall which will require less flexibility, but more strength, for placing their feet on holds which are up high or far to the side.

COLLECT THEM ALL

In advance of the session, tape notes to holds all over the wall; now have the children collect them. They are only allowed to collect one note at a time and they have to place them in a pile some way away from the wall before they can collect another.

Tape the notes in different places with varying levels of difficulty, so that everyone is challenged at their own level.



REBUS

The previous exercise can also be done as a rebus puzzle. This will then be a cooperation exercise where all the notes have either a word or a drawing, and collectively they form the answer to the rebus. The answer could be a song they have to sing or a place with a hidden prize. It's all about being imaginative.

If the rebus puzzle is quite hard, you can number the notes to make it a little easier.



NINJA

A ninja climbs without making a single sound. This is a good exercise for working on precision climbing, and especially precision footwork. You can make this exercise into a competition, where anybody who makes a sound gets a point. Whoever has the least amount of points at the end of the round is the winner.

Instruct the children to look at the footholds as they place their feet.



FOLLOW THE MUSIC

Play different types of music and have the children move to the tunes while they are on the wall. You don't have to lay down any rules for what type of moves they should do. Calm music usually leads to the children focusing on balance, footwork and precise movements, while faster music leads to dynamic and more physical climbing.



Calm music can be used for a good start to the specific warm-up, or alternatively for cooling down after a physical session.



FREEZE WHEN THE MUSIC STOPS

Climb while the music is playing, but as soon as it stops everybody has to freeze where they are and be quiet. If they move or make a sound while the music isn't playing they get a point. Whoever has the least amount of points at the end of the round is the winner.



It's often easier if the music is calm, so you can start with calm music and then make it harder and harder.

NINJA WHEN THE MUSIC STOPS

Let the children climb while the music is playing, but as soon as it stops they have to continue climbing without making a sound. Those who make a sound get a point. Whoever has the least amount of points at the end of the round is the winner.

Vary between slow and fast music to create different movement tempos.



CLIMBING IN PAIRS

Split the group into pairs. Each pair has to hold hands while traversing along the wall using just one hand each. The exercise can also be performed up to the top of the bouldering wall or on top rope, as long as there are suitable routes available. Reaching the finish line will require cooperation and tactics. They have to communicate and agree on who should be moving and lay down a plan for how to reach their target.



Make up a boulder problem that *has* to be climbed in pairs. It's not so hard – just choose holds that are too far apart for one child, but where two children collectively will have enough reach. They can practise the move for a bit and then try to climb the whole problem.



CLIMBING BLIND

Have the children pair up. One child will climb blindfolded, while the other instructs their partner on where to place their hands and feet. It's important that the instructing partner keeps the instructions short, precise and easy to understand. This is a good exercise for training balance and positioning, in addition to being a great exercise for helping the children learn how to cooperate and trust each other.



This exercise can be performed both on the bouldering wall and on top rope, but it can be a good idea to steer clear of the tallest walls. If you're using the bouldering wall it is best to traverse close to the ground.



OBSTACLE COURSE

Use brooms, hula hoops and climbing holds to create an obstacle course on the wall for the children to climb through. They can climb over, under, around and through – the only limiting factor is your own imagination. To make it more challenging, you can tape or mark holds that are off limits. To make it even more challenging, you can tell them it's not allowed to touch any of the obstacles, or you can place objects, such as beanbags or corks, on top of holds that will fall down if they are touched. The exercise is best suited to slabs and vertical walls.



Let the children take part in planning and designing the obstacle course.



SPEED CLIMBING

This is always a hit! A speed climbing competition can be done in many ways, but in our experience it's best to use a short wall with auto-belays – as coach you can then focus on recording the time instead of belaying. Alternatively, the children can climb a boulder problem three times in a row in the shortest time possible. A boulder problem can be climbed so quickly it might be difficult to time it correctly, so it might be better to count the total time for all three attempts. Choose an easy route or boulder problem so that everybody can top it, and encourage the children to improve their own time instead of getting the best overall time.



Ensure good safety routines if using the top rope walls.

MASTERY

As a coach for children and youth groups it's important to create mastery-oriented athletes. As opposed to being result-oriented athletes, they will now be focused on their own progress, mastery and love for the sport, and not just the result of winning a competition or climbing a boulder problem or route before others. To create mastery-oriented environments and athletes it's important to reward personal progress and effort. As a coach you should not just reward the best, but also those who have improved their results compared to last time, or those who decided to give it their all even though they ended up in fifth place.



CANDY CLIMB

The classic game of placing sweets on the holds to make the children even more psyched to climb is perhaps not the healthiest and best game to play every day, but it can be a fun alternative for a Christmas or summer holiday get-together.



Remember, climbing holds aren't clean, so only use sweets with a wrapper!



RELAY RACE

This is a good alternative to a speed climbing competition and it rewards teamwork and cooperation instead of being an individual competition. Divide the group into two teams and line them up five metres from the bouldering wall. Have them run to the wall, climb up to a certain point, jump down and run back to tag the next climber in line. You can be inventive and add in different tasks to be performed during the race.



A relay race should be done on the bouldering wall to eliminate elements of risk from rushing up a top rope wall.

TECHNIQUE EXERCISES

ALL OF THE exercises described in [chapter 1](#) are also suitable for children and youths, they just have to be scaled down to suit shorter climbers with less reach and smaller hands. In addition, in this section we present some new exercises as well as bringing back some familiar ones with a twist to make them more suitable for younger climbers.



BOTTLE TOPS

Tape bottle tops to the soles of the children's shoes so that only the tips of the shoes can be placed on the footholds.

This is a good exercise for training footwork since many children are sloppy when it comes to what part of the shoe they place on the footholds. This is often the result of focusing too much on their hands and the route above instead of looking at their feet as they place them on the footholds.

Perhaps they also do not realise what a disadvantage it is to place something other than the tip of the shoe on the foothold, especially on smaller edges. Therefore, start the exercise by explaining and showing how the feet can twist off the footholds as they move around on the wall unless they have been precise in their foot placements by placing the tips of their shoes on the holds.

This will be especially apparent when traversing or moving sideways using sidepulls, where we twist the body from side to side while moving on the wall.



If you don't have any bottle tops, you can use a slippery tape and tape off the part of the shoe that shouldn't be used.



CLIMB UP AND DOWN

By downclimbing routes after reaching the top the children will be forced to look at their feet when placing them. We recommend pairing the children up, so that they can take turns belaying each other. The exercise can also be done using an auto-belay, but this is not ideal, as the auto-belay will pull them upwards as they are trying to climb down.

Start with easy, vertical routes well within their limit. It's much harder to downclimb than you might think.



MATCHING FEET

Matching every foothold while traversing is a good way to practise this important technique. It's also a good exercise for training precision footwork and balance. Vary the wall angles and have the children experiment with footholds of different sizes and shapes.



Have the children practise different ways of matching their feet: placing one foot next to the other, sliding one foot into the same spot as they remove the other foot, and jump matching.



BABY STEPS

Doing multiple moves with our feet for every move with a hand is a good way to practise precision footwork, and it naturally makes us more aware of the importance of looking down at our feet as we're placing them. For this exercise, have the children do three foot moves for every hand move.



Choose lines on the wall where multiple routes have been set next to or on top of one another. This allows the children to use footholds from a different route if needed.



GIANT LEAPS

Having to place our feet high is common in climbing, especially for shorter climbers, so this is an essential skill to practise. For this exercise, have the children do the exact opposite of the previous exercise – they are to climb a route with as few foot moves as possible, and for every foot move they should try to place their foot as high as possible.



To force the children into keeping their hips close to the wall, have them use only poor holds, like slopers.



QUICKDRAW ON THE HEELS

Attach quickdraws to the heel loops of the children's shoes. Have them climb as calmly as possible so that the quickdraws don't make any sound by bouncing against the wall. This requires both precise footwork and good balance as they move about on the wall.



Give the children some time to figure things out: to begin with, it might be difficult for them to get a feel for how much the quickdraws will swing around.



TENNIS BALL UNDER THE ARM

Have the children climb to the top of the bouldering wall or traverse an entire section of the climbing wall while holding tennis balls under their arms. This puts a significant constraint on the use of their arms, and so is a good exercise for training footwork and balance.



This exercise is best suited to vertical walls and slabs.



GUIDING

For this exercise the children are to guide each other up a route or boulder problem. One climbs, while the other points to which holds the climber is allowed to use. It's easier with free feet – meaning all footholds are allowed – on a bouldering or training wall with a lot of holds. The guide will get a better understanding of movements, and the climber gets to try moves that are different to what they might otherwise have tried.



Use a brush stick or a broom handle to point to holds.

ADD-ON

This is one of the most classic exercises in climbing. One by one, each child adds one move to the boulder problem they are climbing. The first one out defines a starting hold and climbs the first move, then the second one out climbs the first move and adds another move, climbing two moves in total. Third one out climbs the first two moves and adds a third move, and so on. The exercise helps the children remember moves, which is a very important skill in climbing. Additionally, they learn how to make up moves that work for everyone in the group, and this requires an understanding of technique and knowing each other's strengths and weaknesses. The goal is to make up a boulder problem that everyone can do.



Keep the groups relatively small, or there will be too much waiting around.



ELIMINATION

In this exercise the children are tasked with eliminating moves one by one from a boulder problem they have already climbed. Both handholds and footholds can be eliminated. Have them team up in small groups, with each climber removing one hold before taking their turn to climb.



PHOTO: VEGAR HERSTRØM

MAKING UP PROBLEMS

Naturally, as the coach, you should prepare boulder problems that are suited for the session in advance, but why not let the children do it sometimes as well? This works very well in our experience. It's fun for the children to make up problems and they learn a lot from it. Try giving them specific themes, and have them make up problems that everyone in the group can climb. When it's time for the session to begin they can show off their problems and then everyone gets to try. If the problems aren't a good fit for everyone in the group, you can add a few footholds or tweak it slightly so that everyone can benefit from climbing it.

BOULDERING SESSION TIPS

Ida-Sofie Pettersen, a Trondheim Climbing Club coach, shares how she structures the bouldering sessions: 'I usually add different challenging elements to each problem, so that everyone can climb the same problems. Ideally, we start with an easy problem that everyone can manage within a few tries, and then I eliminate a few holds or tell them to not match any holds. This way I can keep the group together without it being too difficult or boring for anyone in the group.'

TRAVERSING

TRAVERSING IS AN excellent way to activate the whole group, and at the same time it's an opportunity to give them tasks for practising different technical elements. To make it into an interesting learning experience we suggest the following exercises.



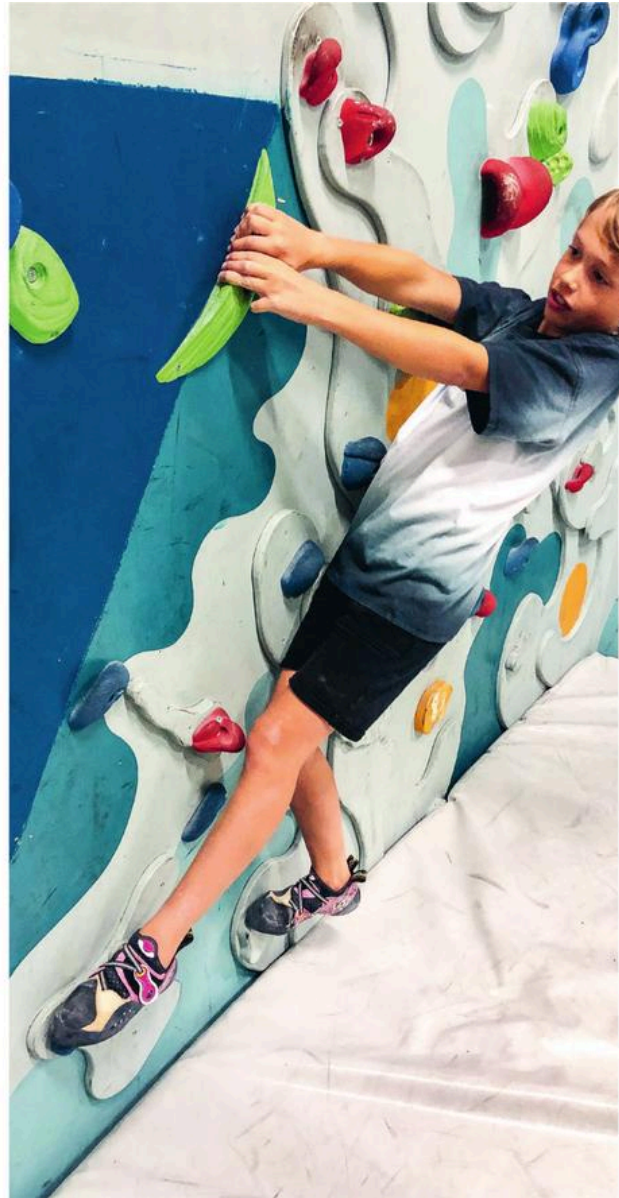
ONE-ARMED BANDIT

For this exercise the children are only allowed to use either their right hand or their left hand. They have to keep their other hand around their back. This is a good exercise for training balance and footwork. The exercise is best suited to slabs or vertical walls with large footholds.



HANDS ON YOUR BACK

Perhaps it's possible to traverse the slab wall without using any hands at all? These kinds of challenges are fun and they are excellent methods for learning footwork and balance. As a coach, you have to find a suitable location for this exercise; it should be a slab with a lot of large holds and volumes big enough to be used as footholds.



SIDEPULLS

Traversing using only sidepulls is a good method for learning balance and positioning. Vertical walls are a good place to start with this exercise. To make it extra challenging, ask the children to try to avoid pinching any of the holds. As we mentioned in [chapter 1](#), not using our thumbs on the holds requires better positioning of our bodies.



DOUBLES

Traversing the whole wall by only doing double dynos, big or small, is a good way for the children to familiarise themselves with dynamic climbing. Many can find this mentally challenging, so it's important to let them figure things out at their own pace. You should demonstrate in advance how to drop the hips slightly down and out from the wall to initiate the movement, before pulling up and in towards the next hold.



ROPE AROUND THE WAIST

Have the children attach a short rope around their waist with a quickdraw at the end to weigh it down; the quickdraw shouldn't be touching the floor when they are climbing. They should now traverse the wall while focusing on maintaining balance and a steady pace. If they move smoothly and correctly the quickdraw will also move smoothly and not swing back and forth.



LET THE CHILDREN FIGURE IT OUT ON THEIR OWN

In our experience, the children will learn a lot more if they figure out a move or a boulder problem on their own. You should therefore give them the tasks without too much instruction in advance; or don't give them too many corrections while they're doing the tasks. You should of course start the session by showing them what to do, and if they're learning a new technique they'll need clear instructions, but then it's important that you take a step back. It's easy to be impatient as a coach and to want to provide too much feedback too soon, because it's so obvious to you what mistakes they are making. But this can diminish their learning experience. Instead, try to give them challenges where they can learn for themselves what works and what doesn't by trial and error. This will also make it easier for them to understand what you mean when you do provide feedback.

STRENGTH TRAINING FOR CHILDREN & YOUTHS

WHETHER CHILDREN AND youths should be doing any strength training has been a subject of discussion for many years, and has been shrouded in myths. Research has now shown quite clearly that strength training is safe, and the effects are important for motor skills development, learning technique, muscular development, power, risk of injury and general health. There are no good arguments for why children and youths should not be doing strength training, and as long as the children are motivated and can receive instructions it's entirely safe to begin with structured strength training for children from the age of five or six.

We should start by defining 'children' and 'youths'. It's hard to put an exact age to these, but in research girls and boys are defined as children up to the ages of 11 and 13 respectively. Youths are defined as the period from 12 to 18 for girls, and 14 to 18 for boys. However, there can be large individual differences, and two 12-year-olds with the same chronological age can have entirely different biological ages. They can also have different training ages, meaning how many years they have been doing structured training. As coaches for children and youths these are important factors we have to consider when planning group training sessions and individual exercise programs.

There will also be significant differences between boys and girls which we have to consider when planning strength training. Throughout puberty, boys will have a stronger response to strength training because of increased levels of testosterone production and therefore increased muscle mass, and the differences in relative strength, meaning strength in relation to body weight, will thereby increase between boys and girls during this period. Girls will benefit from strength training, but will not progress at the same speed as boys, because of hormonal differences related to the production of oestrogen versus testosterone. Where boys during this period can shift their focus towards training more and harder, girls will need a slower progression in terms of load and volume. It's important that we as coaches are aware of this difference, so that we can contribute to individualising goals and controlling the load related to strength training.



WHY SHOULD CHILDREN & YOUTHS TRAIN STRENGTH?

WE WANT TO make one thing entirely clear: performance and results are not the most important reasons for beginning strength training for children and youths. Better results become more achievable if strength training is incorporated into regular training, but what we want to create are healthy, strong robust humans, who, for the rest of their lives, can reap the benefits of the training they did when they were young, and who will continue to live active lives. We can say that strong kids become strong adults, as long as they continue to train throughout their lives.

Starting from a young age our bodies respond positively to strength training. Where at first the effects in children will primarily be linked to neurological adaptation and better execution of movements, the effects through puberty will be directly linked to muscle elasticity and increased muscle mass. Together, this will result in increased muscle strength and an increased force development. This is important for the development of motor skills, for learning sport-specific techniques, and for reducing the risk of injury. All in all, strength training thereby creates robust athletes with long-term performance potential at a lower risk of injury. Additionally, we gain health benefits like increased bone density, reduced levels of fat and reduced obesity, thereby improving overall mental health and lowering the risk of lifestyle-related diseases.

Strength training does not have a negative impact on bone growth zones and will not inhibit growth like some of the myths claim. We should of course be aware of the fact that if the training dosage is too high over extended periods of time this can lead to bone damage in the growth zones, but this would be because of the excessive training dosage, and not the strength training itself. With the right methods we can argue that the risk of injury is reduced through a structured and individually adapted training program, in combination with good load control.



HOW SHOULD CHILDREN & YOUTHS TRAIN STRENGTH?



THE TIMING AND conditions for training different qualities will change throughout the development of children and youths. As mentioned, there can be large differences between children and youths of the same age, but there are still some basic principles we can adhere to.

For children, it's important that training is varied and fun. The focus should be on so-called multi-part exercises, which target larger movements and functions instead of training specific muscle groups. The training should be dynamic, not static, and the exercises should be based around using the arms and legs to carry body weight. The strength training itself should last for 15 to 20 minutes and the exercises should be physically demanding enough for there to be an effect on the nervous system. Climbing is a sport that especially focuses on the upper body, but we must make sure to also include the legs, stomach and back in the training, to stimulate the development of basic motor skills.

Here is a selection of exercises that can be done either individually or in pairs, and either alone or during organised group sessions. It's important that the children can receive instructions, and to begin with the focus should be on learning the technique of the movements. Initially, this shouldn't be a competition about who can do the most, the hardest or the fastest. Make individual adaptations and use deloading where necessary, so that the movements are executed with the proper technique. As the children get older and gain more experience with strength training, the training dosage can increase for each exercise and the internal competitiveness can be used more actively to train more, better and harder.

We recommend adding the selection of exercises as a regular part of the organised training, so that you as a coach can make individual adjustments and contribute to learning of the proper technique. These exercises do not require any additional equipment beyond the kids' own body weight, and they can be added as part of the warm-up or at the end of a session.

GOOD BASE EXERCISES TO BEGIN WITH:

PUSH-UPS

Begin by training the starting position, by keeping a straight and rigid body, without arching the back or bending at the hips. Then move on to lowering the chest deeper towards the floor.

PULL-UPS

Begin hanging from a bar and deload by either using an elastic band or by getting help from a coach so that the lift is controlled and steady.

SIT-UPS

Lie on the ground with knees bent and then lift the upper body up from the floor.

SQUATS

‘Take a seat on the toilet’ is usually a hit with the kids! Have them practise standing in a stable position and squatting down with their heels still in contact with the ground.

HANGING LEG LIFTS

Hang from a bar and lift the knees up to the chest.

RECOMMENDED EXERCISES FOR PAIRS:

WHEELBARROW

One athlete gets down into the push-up position while their partner grabs their ankles and lifts their legs up from the ground. Now walk like a farmer carrying a wheelbarrow forwards, backwards and side to side.

SIT-UPS WITH A CLAP ON TOP

Both athletes lie on their backs feet facing each other with their feet together, and give each other a high five with alternating hands at the top of every sit-up.

PUSH-UPS WITH A CLAP ON TOP

Both athletes face each other in the push-up position, and give each other a high five with alternating hands. Have them vary how far to the side and how high they clap, to challenge the starting position.

ONE-LEGGED BALANCE

Both athletes face each while other standing on one leg, pressing their palms against their partner’s, trying to push each other off balance.

PUSH BACK AND FORTH

Both athletes face each other, pressing their palms against their partner’s. One athlete pushes the other backwards, then they switch.

Early on in puberty we can begin to add new exercises with more rapid movements, directional changes and so-called plyometric exercises. A plyometric exercise is one such as a burpee or a jump squat, where you slow down a movement before exploding back up. This will stimulate the elastic abilities of the muscles and tendons, so that they act like springs and can store and release elastic energy. We also recommend introducing the exercises that are done on the wall, as described in [chapter 2](#). You should avoid doing the exercises on small, positive holds, and you should be prepared to make adjustments for the group you are coaching. Additionally, we recommend:

HAND SLAP PUSH-UPS

Both athletes face each other in the push-up position. The goal is to slap the ‘opponent’s’ hands without getting slapped themselves.

PALM WRESTLING

Palms should be touching at all times, and the goal is to wrestle the ‘opponent’ out of a marked area or down to the ground.

BURPEES

Start in the push-up position. Do a quick push-up, then jump with arms above the head before returning to the push-up position.

QUICK PULL-UPS

Do pull-ups as quickly as possible, preferably using an elastic band to deload.

JUMP TO TWO-HANDED LOCK-OFFS

Jump from the ground to a bar, landing with the elbows bent at different angles. This can be done in combination with burpees.

TIC-TAC

Hang from two good holds. Choose three holds at different heights on each side of the body and place a foot on each hold, one by one, while returning to the starting position every time.

BODY WEIGHT AND LOADING

Body weight increases during puberty through an increase in muscle mass and fat, and this increase is different for every athlete. It's important that we maintain a dialogue with our athletes regarding this, as an increase in body weight will increase the load on the muscles, tendons and joints. In periods of rapid growth, we have to individualise the training and perhaps reduce the load for a period, so that the body can adapt to the new weight. This is especially important for finger strength training, where even just a few kilos of increased body weight can result in a significant increase in load on finger joints and on the muscle-tendon apparatus of the fingers.



Jump to lock-off.



Tic-tac.



Hand slap push-ups can be done both as a warm-up and as strength training. When the athletes can do the exercise with proper technique and in control, you can add competitive elements like trying to get the most slaps possible in 30 seconds.

During puberty, strength training will gradually start to resemble the strength training we do as adults, focusing more on muscle growth and increased strength in more specific movement patterns. We should carry with us the principles of individualising the training to everyone's level and training age, and be conscious of the fact that the way to go is to train the individual in different ways, adapted to their needs and development level. Boys, especially, will increase their muscle mass during puberty, and many of the gender-related differences in strength are related to the differences in muscle mass. Increased muscle mass and further neurological development will

collectively lead to an increase in force development and stronger athletes. A thoroughly fit and strong youth will be at a lower risk of injury and be at an advantage when it comes to enduring a gradual increase in training load and learning increasingly difficult techniques. It's natural to actively use the exercises for strength and power on the wall in the training. Additionally, we recommend beginning with some isolated exercises.



DEADHANGS

Choose different holds and grip positions, for example half crimp on deep edges, three finger pockets, or an open-handed grip on slopy half domes. Maintain an 'active' hanging position, meaning the shoulders are pulled down and the elbows are angled in towards the centre. Hang for 15 to 20 seconds in this position with a margin of 3 to 5 seconds per hang. Repeat three to four sets with a 2- to 3-minute rest in between.

APPENDIX TO FINGER TRAINING

In *The Climbing Bible* we categorically discouraged deadhang training for young climbers. Here we want to modify this statement to make it slightly more nuanced: we still recommend being very careful with specific finger training for younger athletes whose fingers have not yet finished growing, but there are methods that open up the possibility for conducting finger training in a safe and controlled manner.

If we were to discourage deadhangs, we would also have to discourage climbing and bouldering, as the load on the fingers is lower for deadhang training conducted in this way than it is for regular climbing and bouldering sessions.

Controlled strength training for the fingers, incorporated into a sensible training programme, will contribute to an increase in finger strength through safer training methods than hard, finger-intensive climbing and bouldering sessions. In this way, the training can help to reduce the risk of sustaining finger injuries at a later stage. Research findings from other sports also show that structured strength training helps to reduce the risk of injury.

We still discourage campus board training and a singular focus on finger-intensive climbing and bouldering, and we recommend that young climbers vary their training to become as well-rounded and complete as possible. We also recommend that athletes, coaches and parents take any finger-related symptoms seriously, and to undergo medical examination if necessary, so that the training can be adjusted.



ONE-HANDED LOCK-OFFS

As with deadhangs, we can use hang time as a target for the training. This way we can deload as necessary to ensure the lock-off is executed in a technically correct way and in control, and still reap the benefits of strength training. We recommend training at different angles in the elbow, from 90 degrees to almost straight, with a hang time of up to 15 seconds, and a 3- to 5-second margin for every hang. Repeat three to four sets with a 2- to 3-minute rest in between.



PHOTO: JARL GÅSVÆR

Sigrid Baumberger and Ida Baardsen climbing Svolværgeita, Lofoten, Norway



Remember, it's OK to take a break to just enjoy life and the surroundings after a long day out, or after a hard training session. Here's Kasper Fjeldstad Christophersen taking a breather during a round of exploring in the magical forest of Fontainebleau, France.